A depth-one closed form for Apéry's $\zeta(3)$ numerators

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Abstract

We prove the closed form $b_n = \sum_{k=0}^{n} \binom{n}{k}^2 \binom{n+k}{k}^2 \big(2H_n^{(3)} - H_k^{(3)}\big)$ for the second solution of Apéry's $\zeta(3)$ recurrence, where $H_m^{(3)} = \sum_{j=1}^{m} j^{-3}$. The weight $2H_n^{(3)} - H_k^{(3)}$ is of harmonic depth one and is the unique such weight; it arises as the third Taylor coefficient of an explicit Γ-deformation of the Apéry summand. The proof consists of two rational certificates and a two-term induction, and has been formalized in Lean 4.

1 Statement

For $n, k \in \mathbb{N}$ put

$$A(n,k) = \binom{n}{k}^2 \binom{n+k}{k}^2, \qquad a_n = \sum_{k=0}^{n} A(n,k), \qquad H_m^{(3)} = \sum_{j=1}^{m} \frac{1}{j^3}.$$

Let L denote Apéry's difference operator,

$$(\mathrm{L}\,u)_n = (n+1)^3 u_{n+1} - (2n+1)(17n^2+17n+5)\,u_n + n^3 u_{n-1},$$

so that $\mathrm{L}\,a = 0$, and let (b_n) be the second solution: $\mathrm{L}\,b = 0$, $b_0 = 0$, $b_1 = 6$; classically $b_n/a_n \to \zeta(3)$.

Theorem 1. *For all $n \geq 0$,*

$$b_n = \sum_{k=0}^{n} \binom{n}{k}^2 \binom{n+k}{k}^2 \big(2H_n^{(3)} - H_k^{(3)}\big).$$

Remark 1. The representation is canonical. Writing $S_L(n) = \sum_k A(n,k) H^{(3)}_{L(n,k)}$ for a linear form L, the four sequences $S_n, S_k, S_{n-k}, S_{n+k}$ are $\mathbb{Q}$-linearly independent (a rank computation on $n \leq 15$ witnesses this), and Theorem 1 says $b = 2S_n - S_k$. Thus the weight 3, the depth 1, and the coefficients $(2, -1)$ are invariants of b_n, not choices. By contrast, Apéry's classical representation carries the weight $H_n^{(3)} + \sum_{m=1}^{k} \frac{(-1)^{m-1}}{2m^3\binom{n}{m}\binom{n+m}{m}}$, whose second term is not a harmonic form.

Remark 2. The weight is produced, not guessed, by a deformation. With $\Pi_j(t) = \prod_{i=1}^{t}(1 + j\varepsilon/i)$ set

$$A_\varepsilon(n,k) = A(n,k)\,\Pi_1(n)^6 \Pi_2(n)^{-6} \Pi_3(n)^2\,\Pi_1(k)^{-3} \Pi_2(k)^3 \Pi_3(k)^{-1}.$$

The exponent vectors $u = (6, -6, 2)$, $v = (-3, 3, -1)$ are the unique (up to scale) solutions of $e_1 = e_2 = 0$ for the power sums $e_m(c) = \sum_j c_j j^m$ over the three shift points $j = 1, 2, 3$;

consequently $\log(A_\varepsilon/A) = \varepsilon^3 L_3 + O(\varepsilon^4)$ with $L_3 = \frac{1}{3}\big(e_3(u)H_n^{(3)} + e_3(v)H_k^{(3)}\big) = 2\big(2H_n^{(3)} - H_k^{(3)}\big)$, and the first two Taylor coefficients of $\sum_k A_\varepsilon$ vanish *termwise*. Theorem 1 is then the statement $b_n = \frac{1}{2}[\varepsilon^3]\sum_k A_\varepsilon(n,k)$: the closed form is the primitive third-order coefficient of one deformed hypergeometric family.

2 The certificates

All identities below are between rational numbers; $n \geq 1$ throughout. Define the polynomials

$$\begin{aligned}
N(n,k) &= 4k^4(2n+1)\big(2k^2 - 3k - 4n(n+1)\big),\\
P(n,k) &= k^5 - 2(n^2+n+1)k^4 + (2n^2+2n+1)k^3 + 2n^2(n+1)^2k^2\\
&\qquad - 3n^2(n+1)^2k - 4n^3(n+1)^3,\\
M(n,k) &= 4k(2n+1)\,P(n,k),
\end{aligned}$$

the binomial shell

$$X(n,k) = \binom{n+1}{k}^2\binom{n+k-1}{k}^2,$$

and the two certificate sequences

$$G(n,k) = \frac{N(n,k)\,X(n,k)}{n^2(n+1)^2}, \qquad Y(n,k) = \frac{M(n,k)\,X(n,k)}{n^4(n+1)^4}.$$

Note that $X(n,k) = 0$ for $k \geq n+2$, and $k \mid N$, $k \mid M$, so

$$G(n,0) = Y(n,0) = 0, \qquad G(n,k) = Y(n,k) = 0 \quad (k \geq n+2). \tag{1}$$

Lemma 1. *For $n \geq 1$ and all $k \geq 0$,*

$$(n+1)^3A(n+1,k) - (2n+1)(17n^2+17n+5)A(n,k) + n^3A(n-1,k) = G(n,k+1) - G(n,k).$$

Lemma 2. *For $n \geq 1$ and all $k \geq 0$,*

$$\frac{G(n,k+1)}{(k+1)^3} + 2A(n+1,k) - 2A(n-1,k) = Y(n,k+1) - Y(n,k).$$

Proof of Lemmas 1 and 2. For $k \geq n+2$ every term vanishes, by (1) and $A(n\pm1,k) = A(n,k) = 0$. Let $0 \leq k \leq n+1$, so $X(n,k) > 0$. From $\binom{m}{k}/\binom{m-1}{k} = \frac{m}{m-k}$ and $\binom{m}{k+1}/\binom{m}{k} = \frac{m-k}{k+1}$ one obtains the four ratio identities

$$\frac{A(n,k)}{X(n,k)} = \frac{(n{+}1{-}k)^2(n{+}k)^2}{n^2(n{+}1)^2}, \quad \frac{A(n{+}1,k)}{X(n,k)} = \frac{(n{+}k)^2(n{+}k{+}1)^2}{n^2(n{+}1)^2}, \quad \frac{A(n{-}1,k)}{X(n,k)} = \frac{(n{-}k)^2(n{+}1{-}k)^2}{n^2(n{+}1)^2},$$

$$\frac{X(n,k+1)}{X(n,k)} = \frac{(n{+}1{-}k)^2(n{+}k)^2}{(k+1)^4}$$

(valid as stated for $0 \leq k \leq n+1$: whenever a numerator binomial vanishes, so does the corresponding polynomial factor). Substituting these and multiplying by $n^2(n+1)^2(k+1)^4/X(n,k)$ (resp. $n^4(n+1)^4(k+1)^4/X(n,k)$), the two lemmas become the polynomial identities in $\mathbb{Z}[n,k]$

$$\begin{aligned}
(k{+}1)^4\Big[&(n{+}1)^3(n{+}k)^2(n{+}k{+}1)^2 - (2n{+}1)(17n^2{+}17n{+}5)(n{+}1{-}k)^2(n{+}k)^2\\
&+ n^3(n{-}k)^2(n{+}1{-}k)^2\Big] = N(n,k{+}1)(n{+}1{-}k)^2(n{+}k)^2 - N(n,k)(k{+}1)^4,
\end{aligned} \tag{2}$$

$$n^2(n{+}1)^2\Big[\widetilde{N}(n,k)(n{+}1{-}k)^2(n{+}k)^2+2(k{+}1)^4\big((n{+}k)^2(n{+}k{+}1)^2-(n{-}k)^2(n{+}1{-}k)^2\big)\Big]$$
$$=M(n,k{+}1)(n{+}1{-}k)^2(n{+}k)^2-M(n,k)(k{+}1)^4, \quad (3)$$

where $\widetilde{N}(n,k)=N(n,k+1)/(k+1)^3=4(k{+}1)(2n{+}1)\big(2(k{+}1)^2-3(k{+}1)-4n(n{+}1)\big)$. Both are verified by expansion. $\square$

3 Proof of the theorem

Write $W(n,k)=2H_n^{(3)}-H_k^{(3)}$ and $c_n=\sum_{k=0}^n A(n,k)W(n,k)$; since $A(n,k)=0$ for $k>n$, all sums below may be taken over $0\le k\le n+1$.

Proposition 1. $\mathrm{L}\,c=0$ *for* $n\ge 1$.

Proof. From $W(n\pm 1,k)-W(n,k)=\pm 2/(n{+}\frac{1}{2}\pm\frac{1}{2})^3$, i.e. $W(n{+}1,k)=W(n,k)+2(n{+}1)^{-3}$ and $W(n{-}1,k)=W(n,k)-2n^{-3}$,

$$(\mathrm{L}\,c)_n=\sum_{k=0}^{n+1}\Big[\big(\mathrm{L}\,A(\,\cdot\,,k)\big)_n\,W(n,k)+2A(n{+}1,k)-2A(n{-}1,k)\Big].$$

By Lemma 1 and summation by parts — using $W(n,k)-W(n,k+1)=(k+1)^{-3}$ and the boundary values (1) —

$$\sum_{k=0}^{n+1}\big(G(n,k{+}1)-G(n,k)\big)W(n,k)=\sum_{k=0}^{n+1}\frac{G(n,k{+}1)}{(k+1)^3}.$$

Hence, by Lemma 2 and telescoping,

$$(\mathrm{L}\,c)_n=\sum_{k=0}^{n+1}\big(Y(n,k{+}1)-Y(n,k)\big)=Y(n,n{+}2)-Y(n,0)=0. \qquad\square$$

Proof of Theorem 1. $c_0=0=b_0$ and $c_1=A(1,0)\cdot 2+A(1,1)\cdot 1=2+4=6=b_1$. Since $(n+1)^3\neq 0$, the relation $\mathrm{L}\,u=0$ determines u_{n+1} from (u_{n-1},u_n); by Proposition 1 and induction, $c_n=b_n$ for all n. $\square$

Remark 3. Only the telescoped value $Y(n,n+2)-Y(n,0)=0$ is used, but the certificate Y encodes finer information: its rational continuation satisfies $Y(n,n+1)=-2\binom{2n+2}{n+1}^2$, from $M(n,n{+}1)=-8(2n{+}1)^2n^2(n{+}1)^4$, which is precisely the $k=n{+}1$ term of $2a_{n+1}$ — the summand the range $k\le n$ cannot see.

Remark 4. The proof is a template. Whenever a sequence $B_n=\sum_k S(n,k)w(n,k)$ has a summand S with a first-order (Zeilberger) certificate for its own row $\sum_k S$, and a weight w whose letters have arguments among $\{n,k\}$ only, the shifts $w(n\pm 1,k)-w(n,k)$, $w(n,k+1)-w(n,k)$ are explicit rationals, the operator splits as above, and the recurrence for B reduces to a single Gosper problem. The Franel case ($S=\binom{n}{k}^3$, $w=\frac{1}{4}H_k^{(2)}+\frac{3}{4}H_k(H_k-H_{n-k})$) and its $\sum\binom{n}{k}^4$ analogue are within immediate reach of the same argument.

Remark 5. A complete, sorry-free Lean 4 formalization of Theorem 1 (definitions over $\mathbb{Q}$, the two certificate identities, and the induction) accompanies this note in the `ZetaLucas` library, file `AperyClosedForm.lean`. The polynomial identities (2)–(3) were additionally verified cellwise in exact rational arithmetic for $n\le 14$, all $k\le n+3$.

References

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